

Lab 8: Computational (Quadratic) Optimization

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Objective: In this lab, you will get familiarized with computational quadratic optimization by modeling real-life standard problems. To solve quadratic problems, use the existing Pyomo framework with **cbc** solver or **qpsolvers** or **gurobipy** instead of GLPK solver, as the nature of the objective function is quadratic (or can be non-convex) in nature.

Instructions: Below are some instructions. Please go through them carefully:

- Along with the .ipynb file, you must submit a report (**.pdf, not a photo-captured image** as .pdf file) file that answers all the questions from lab 8. You can also use the markdown option to create a report within the .ipynb file itself.
- Also, explicitly mention the assumptions used throughout your modeling technique.
- Use Pyomo, Numpy, qpsolvers, gurobipy and other library documentation if you need help.
- An introductory ready-to-use Pyomo framework along with instructions is available as an [Introduction to Data Handling and Quadratic Optimization](#) to solve optimization problems for a given data set. Use this .ipynb file(s), save it to your iitb Gdrive, and understand how to handle and utilize the given dataset for optimization.
- Your task is to model the scenarios given below as optimization formulation and report your answers to questions.
- Use the traditional approach to name your files for submission:
 - **ROLLNUMBER_IE507_Lab11_Submission.ipynb** (can be multiple files)
 - **ROLLNUMBER_IE507_Lab11_Report.pdf**

Quadratic Programming

An optimization formulation of the type (1) below is called a quadratic program due to a quadratic objective function and linear constraints.

$$\begin{aligned} \min_x &: \frac{1}{2}x^T Px + q^T x + s \\ \text{subject to :} & \\ & Gx \leq h \end{aligned}$$

where, $x \in \mathcal{R}^n$ is the decision variables. Further, $P \in \mathcal{R}^{n \times n}$, $s \in \mathcal{R}$, $G \in \mathcal{R}^{m \times n}$ where n is the number of decision variables and m is the number of constraints in the system.

We stick to the case when matrix P is positive definite; then the objective function $\frac{1}{2}x^T Px + q^T x + s$ is convex in x and hence has a unique global minimum. Also, recall that the positive definiteness of P is equivalent to all its eigenvalues being positive.

Convexity is a geometric aspect and convex functions have ‘curvature’, whose orientation is in a certain sense ‘ordered’; plot a nonconvex function, say $\sin x$ and $\dots$. We skip zeroth and first order characterisations of convexity. Also since, s independent to decision variable x , so we minimizing problem (1) above is equivalent to minimizing objective function $\frac{1}{2}x^T Px + q^T x$.

A (computational) consequence of this convexity is that any local minima is a global minima; so, any descent algorithm will eventual yield global minima.

Some further background: Recall, that if a Linear Program has an optimal value, then its dual is a LP and has an optimal value and both values are the same. Now, one can ask a question, about the class of

optimization problems with such results; turns out that for finite dimensional optimization problems, convex programs have this property (under mild regularity conditions). Convex quadratic programs, being convex programs, have this property; we skip the details.

Converting non-linear program to linear program

Consider a standard example of absolute value given below:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{otherwise} . \end{cases} \quad (1)$$

It is known that the absolute function is non-linear and can be converted into a linear program with some manipulations. We will now consider the absolute value function appearing in the objective function in a standard optimization problem **(OP)**:

$$\begin{aligned} \min \quad & |x_1| + |x_2| + |x_3| + |x_4| + |x_5| \\ \text{s.t.} \quad & 85x_1 + 92x_2 + 45x_3 + 27x_4 + 31x_5 \geq 1 \\ & 92x_1 + 54x_2 + 22x_3 + 20x_4 + 7x_5 \geq 1 \\ & 96x_1 + 67x_2 + 29x_3 + 20x_4 + 11x_5 \geq 1 \\ & -91x_1 - 57x_2 - 33x_3 - 23x_4 - 12x_5 \geq 1 \\ & -99x_1 - 75x_2 - 26x_3 - 24x_4 - 41x_5 \geq 1 \\ & -98x_1 - 99x_2 - 57x_3 - 45x_4 - 65x_5 \geq 1 \end{aligned}$$

In this optimization problem **(OP)**, we wish to minimize the sum of absolute values of the decision variables. Such problems are helpful when the decision variables might take both positive and negative values, and we want to optimize the magnitudes of the decision variables. The following two approaches traditionally convert the above **(OP)** into a standard linear program.

Approach 1: OP1

We can use a substitution of the absolute function values of the variables to new variables as $u_i = |x_i|$, $\forall i \in \{1, \dots, 5\}$. With this substitution, note that we have $u_i \geq 0$, $\forall i \in \{1, \dots, 5\}$. Also, the following inequalities are satisfied: $x_i \leq u_i$, $-x_i \leq u_i$, $\forall i \in \{1, \dots, 5\}$. Thus we can transform the problem **(OP)** as the following optimization problem **(OP1)**, where the absolute values of x_i are replaced with u_i and constraints related to the new variables u_i are introduced. Thus we have the problem **(OP1)**:

$$\begin{aligned} \min \quad & u_1 + u_2 + u_3 + u_4 + u_5 \\ \text{s.t.} \quad & 85x_1 + 92x_2 + 45x_3 + 27x_4 + 31x_5 \geq 1 \\ & 92x_1 + 54x_2 + 22x_3 + 20x_4 + 7x_5 \geq 1 \\ & 96x_1 + 67x_2 + 29x_3 + 20x_4 + 11x_5 \geq 1 \\ & -91x_1 - 57x_2 - 33x_3 - 23x_4 - 12x_5 \geq 1 \\ & -99x_1 - 75x_2 - 26x_3 - 24x_4 - 41x_5 \geq 1 \\ & -98x_1 - 99x_2 - 57x_3 - 45x_4 - 65x_5 \geq 1 \\ & u_i \geq x_i, \forall i \in \{1, \dots, 5\} \\ & u_i \geq -x_i, \forall i \in \{1, \dots, 5\} \\ & u_i \geq 0, \forall i \in \{1, \dots, 5\}. \end{aligned} \quad (2)$$

Note that the number of decision variables in **(OP1)** is twice the number of decision variables in **(OP)**. However the objective function and the constraints in **(OP1)** are now linear, and thus **(OP1)** is a linear program.

Question 1. Objective function with absolute value function

Consider the linear program formulation for **(OP)** non-linear program into **(OP1)** and **(OP2)**.

1. Create python DataFrame coeffOP1 for **(OP1)**, containing the coefficients of the objective function and constraints. Further, create a Pyomo Framework using this created DataFrame for **(OP1)**.

Question 2. Best Fit: Minimizing Least Square Estimates

You are an asset manager for a large cooperation bank; over the year, the market has observed volatility in interest rates of Sovereign Bond issued by the central bank. A data set for [interest rate](#) has been provided to you for the last few years. Your team is interested in forecasting the central bank's future interest rate for these bonds based on historical information. Two factors, $\tilde{R}$ and R , significantly contribute to determining interest rates. $\tilde{R}$ represents the export-import ratio, and R is the rating from the international rating agency. So, as a rule of thumb, the bank feels that a linear model of the form $r = b_0 + b_1\tilde{R} + b_2R$ would reasonably perform well for estimation.

You would like to select the “best” linear model in some sense. If you knew the three parameters b_0 , b_1 , and b_2 , the observations $i \in \{1, 2, \dots, 20\}$ in the data would each provide a forecast of the interest rate as follows:

$$\hat{r}_i = b_0 + b_1\tilde{R} + b_2R$$

However, since b_0 , b_1 , and b_2 cannot, in general, be chosen so that the actual interest rate r_i is precisely equal to the forecast interest rate $\hat{r}_i$ for all observations, you would like to minimize the residuals $\epsilon_i = r_i - \hat{r}_i$.

1. Write a general optimization problem using the actual objective provided in the description in matrix notations. You may assume that $b_0 \geq 0$ and b_1, b_2 do not have bounds on them. Is the optimization problem linear? Explain.
2. Solve the above formulation using the Pyomo/qpsolver framework and report your optimal solution. (**Hint:** residuals (ϵ_i) might cancel each other when added, so one may consider the square of residuals.)
3. Assume you are interested in minimizing the absolute deviation between actual and predicted interest rates. How will you modify the above formulation? It is known that such an approach will result in a non-linear program. Use **(OP1)** approach to determine decision variables.
4. Plot your given data along with the predicted data using your decision variables from mean square deviation and absolute deviation. Also, solve your modified formulation to obtain model parameters.

OP2 We have one more approach to handle $|\cdot|$ in objective function. In this approach, we will use some fundamental properties of real numbers. Indeed, for any real number x , we can write $x = a - b$ where $a \geq 0, b \geq 0$. Hence, we can substitute $|x| = |a - b|$, $a \geq 0, b \geq 0$.

Now we will consider another interesting property of real numbers: $|u - v| = u + v \iff u \geq 0, v \geq 0, uv = 0$.

Using this property we can write $|x| = a + b$, $a \geq 0, b \geq 0$. Note that this replacement will imply that $ab = 0$.

Thus we can transform the optimization problem **(OP)** into a new optimization problem **(OP2)** where $x_i = a_i - b_i$, and $|x_i| = a_i + b_i$, $\forall i \in \{1, \dots, 5\}$ where $a_i \geq 0, b_i \geq 0, \forall i \in \{1, \dots, 5\}$. Thus we have the new optimization problem **(OP2)**:

$$\begin{aligned}
& \min (a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) + (a_4 + b_4) + (a_5 + b_5) \\
\text{s.t. } & 85(a_1 - b_1) + 92(a_2 - b_2) + 45(a_3 - b_3) + 27(a_4 - b_4) + 31(a_5 - b_5) \geq 1 \\
& 92(a_1 - b_1) + 54(a_2 - b_2) + 22(a_3 - b_3) + 20(a_4 - b_4) + 7(a_5 - b_5) \geq 1 \\
& 96(a_1 - b_1) + 67(a_2 - b_2) + 29(a_3 - b_3) + 20(a_4 - b_4) + 11(a_5 - b_5) \geq 1 \\
& -91(a_1 - b_1) - 57(a_2 - b_2) - 33(a_3 - b_3) - 23(a_4 - b_4) - 12(a_5 - b_5) \geq 1 \\
& -99(a_1 - b_1) - 75(a_2 - b_2) - 26(a_3 - b_3) - 24(a_4 - b_4) - 41(a_5 - b_5) \geq 1 \\
& -98(a_1 - b_1) - 99(a_2 - b_2) - 57(a_3 - b_3) - 45(a_4 - b_4) - 65(a_5 - b_5) \geq 1 \\
& a_i \geq 0, b_i \geq 0, \forall i \in \{1, \dots, 5\}.
\end{aligned}$$

5. Now modify the optimization problem you obtained from absolute deviation and use **OP2** to convert your optimization problem. Also, solve the modified optimization problem.
6. Both **(OP1)** and **(OP2)** have same number of constraints. Comment on the activity of constraints of one formulation compared to other formulation.